

Solutions for Problem Set 7

Mathematics for Social Scientists

EXERCISE 1. Write the following system of linear equations as a Gaussian tableau, and then put the tableau in *reduced* row echelon form using elementary row operations.

$$x_1 - x_2 - x_3 = -3$$

$$2x_1 + 6x_3 = 4$$

$$-2x_1 + 2x_2 + x_3 = 5$$

Gaussian tableau: $\left(\begin{array}{ccc|c} 1 & -1 & -1 & -3 \\ 2 & 0 & 6 & 4 \\ -2 & 2 & 1 & 5 \end{array} \right)$, Reduced row echolon form: $\left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right)$

EXERCISE 2. For the following SLEs, determine the rank, extended rank, and the set of solutions.¹

(i)

$$3x_1 - x_2 + 8x_3 = 2$$

$$x_1 + 5x_2 + x_3 = 0$$

$$x_1 - 11x_2 + 6x_3 = 3$$

- Rank: 2
- Extended Rank: 3
- Set of Solutions: No solution, $L = \emptyset$

(ii)

$$x_1 + 2x_2 = 1$$

$$2x_1 + 2x_2 + x_3 = 5$$

$$3x_1 + 7x_2 + x_3 = 3$$

¹Write the associated Gaussian tableau, Reduce the extended matrix to row echolon form using elementary row operations, determine the rank and then find the set of solutions L .

- Rank:3
- Extended Rank:3
- Set of Solutions:Unique Solution, $|L| = 1$

(iii)

$$7x_1 + 2x_2 + x_3 = 1$$

$$x_1 + 5x_2 + x_3 = 4$$

$$3x_1 + 4x_2 + x_3 = 3$$

- Rank: 2
- Extended Rank: 3
- Set of Solutions: Infinitely many solutions

EXERCISE 3. Construct the 3 x 4 matrix $\mathbf{A} = (a_{ij})_{3 \times 4}$ where $a_{ij} = 3i - 2j$.

$$\begin{pmatrix} 1 & -1 & -3 & -5 \\ 4 & 2 & 0 & -2 \\ 7 & 5 & 3 & 1 \end{pmatrix}$$

EXERCISE 4. (*Essential Mathematics for Economic Analysis (5th edition)*, Knut Sydsaeter and Peter Hammond, Section 15.2, Question 3 with a little twist)

Determine for what values of u and v the following equality holds²:

$$\begin{pmatrix} (1-u)^2 & v^2 & 3 \\ v & 2u & 5 \\ 6 & u & -1 \end{pmatrix} = \begin{pmatrix} 4 & v & 6 \\ 4 & -3v & v+5 \\ u & u-v & -1 \end{pmatrix}^T$$

$$u = 3, v = -2$$

EXERCISE 5. (*Essential Mathematics for Economic Analysis (5th edition)*, Knut Sydsaeter and Peter Hammond, Section 15.5, Question 4)

²Here, the T means tranpose

Determine the values of a for which the matrix $\begin{pmatrix} a & a^2 - 1 & -3 \\ a + 1 & 2 & a^2 + 4 \\ -3 & 4a & -1 \end{pmatrix}$ is symmetric.

$$a = 2$$

EXERCISE 6. Find the inverses of the following matrices. You do not have to show that they are invertible.

(i)

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$$

$$\begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}$$

(ii)

$$B = \begin{pmatrix} 0 & 1 & -4 \\ 1 & 2 & -1 \\ 1 & 1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 5 & -6 & 7 \\ -3 & 4 & -4 \\ -1 & 1 & -1 \end{pmatrix}$$